

# Title of Essay

| Library      |                   | # Type<br>defs | # Generic<br>defs | Type Definitions |            |            |            |           | Recursive<br>variances | Unnecess.<br>wildcards | Over-spezif.<br>methods |
|--------------|-------------------|----------------|-------------------|------------------|------------|------------|------------|-----------|------------------------|------------------------|-------------------------|
|              |                   |                |                   | invar.           | variant    | cov.       | contrav.   | biv.      |                        |                        |                         |
| Guava        | classes           | 209            | 106               | 81%              | 19%        | 11%        | 8%         | 0%        | 9%                     | 51%                    | 20%                     |
|              | interfaces        | 37             | 28                | 43%              | 57%        | 36%        | 18%        | 4%        | 18%                    | 38%                    | 5%                      |
|              | total             | 246            | 134               | 73%              | 27%        | 16%        | 10%        | 1%        | 11%                    | 51%                    | 20%                     |
| Java         | classes           | 5565           | 114               | 60%              | 40%        | 21%        | 15%        | 4%        | 11%                    | 12%                    | 6%                      |
|              | interfaces        | 1714           | 48                | 40%              | 60%        | 38%        | 19%        | 4%        | 22%                    | 12%                    | 17%                     |
|              | total             | 7279           | 162               | 54%              | 46%        | 26%        | 16%        | 4%        | 14%                    | 12%                    | 8%                      |
| JPaul        | classes           | 82             | 70                | 71%              | 29%        | 10%        | 13%        | 6%        | 18%                    | 84%                    | 19%                     |
|              | interfaces        | 11             | 11                | 73%              | 27%        | 0%         | 27%        | 0%        | 9%                     | 0%                     | 0%                      |
|              | total             | 93             | 81                | 72%              | 28%        | 9%         | 15%        | 5%        | 17%                    | 84%                    | 19%                     |
| JScience     | classes           | 73             | 28                | 75%              | 25%        | 0%         | 0%         | 25%       | 83%                    | 90%                    | 24%                     |
|              | interfaces        | 52             | 12                | 50%              | 50%        | 8%         | 8%         | 33%       | 14%                    | 100%                   | 11%                     |
|              | total             | 125            | 40                | 68%              | 32%        | 2%         | 2%         | 28%       | 60%                    | 91%                    | 24%                     |
| Apache       | classes           | 243            | 204               | 67%              | 33%        | 13%        | 19%        | 1%        | 4%                     | 63%                    | 17%                     |
|              | interfaces        | 26             | 25                | 56%              | 44%        | 28%        | 16%        | 0%        | 24%                    | 20%                    | 0%                      |
|              | total             | 269            | 229               | 66%              | 34%        | 14%        | 19%        | 1%        | 6%                     | 62%                    | 17%                     |
| Trove        | classes           | 25             | 8                 | 62%              | 38%        | 12%        | 25%        | 0%        | 33%                    | 0%                     | 62%                     |
|              | interfaces        | 9              | 5                 | 0%               | 100%       | 20%        | 80%        | 0%        | 0%                     | 0%                     | 0%                      |
|              | total             | 34             | 13                | 38%              | 62%        | 15%        | 46%        | 0%        | 20%                    | 0%                     | 62%                     |
| <b>Total</b> | <b>classes</b>    | <b>6197</b>    | <b>530</b>        | <b>69%</b>       | <b>31%</b> | <b>13%</b> | <b>14%</b> | <b>4%</b> | <b>12%</b>             | <b>42%</b>             | <b>14%</b>              |
|              | <b>interfaces</b> | <b>1849</b>    | <b>129</b>        | <b>46%</b>       | <b>54%</b> | <b>29%</b> | <b>20%</b> | <b>5%</b> | <b>19%</b>             | <b>19%</b>             | <b>15%</b>              |
|              | <b>total</b>      | <b>8046</b>    | <b>659</b>        | <b>64%</b>       | <b>36%</b> | <b>16%</b> | <b>15%</b> | <b>4%</b> | <b>13%</b>             | <b>40%</b>             | <b>14%</b>              |